

B.Eng Thesis Project Plan

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Document Overview

This document presents the project plan for a Bachelor of Engineering thesis, conducted in collaboration with the company Everllence, focused on extending an existing CAN bus VHDL module to support the Flexible Data rate (FD) protocol ([ISO 11898-1:2024](#)). The project runs from 23-02-2026 to 07-06-2026 (15 weeks) and will deliver a thesis report, VHDL implementation, and a verification suite (VHDL + Python). The plan outlines the objectives, timeline, deliverables, and key risks for the project.

Key Stakeholders

- **Student:** Mads Richardt (DTU B.Eng candidate, s224948, mads.richardt@gmail.com)
 - **DTU Advisor:** Edward Alexandru Todirica (Associate Professor, eato@dtu.dk)
 - **Company Advisor:** Fredrik Kristensen (Everllence, fredrik.kristensen@man-es.com)
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Project Brief (from Company Advisor)

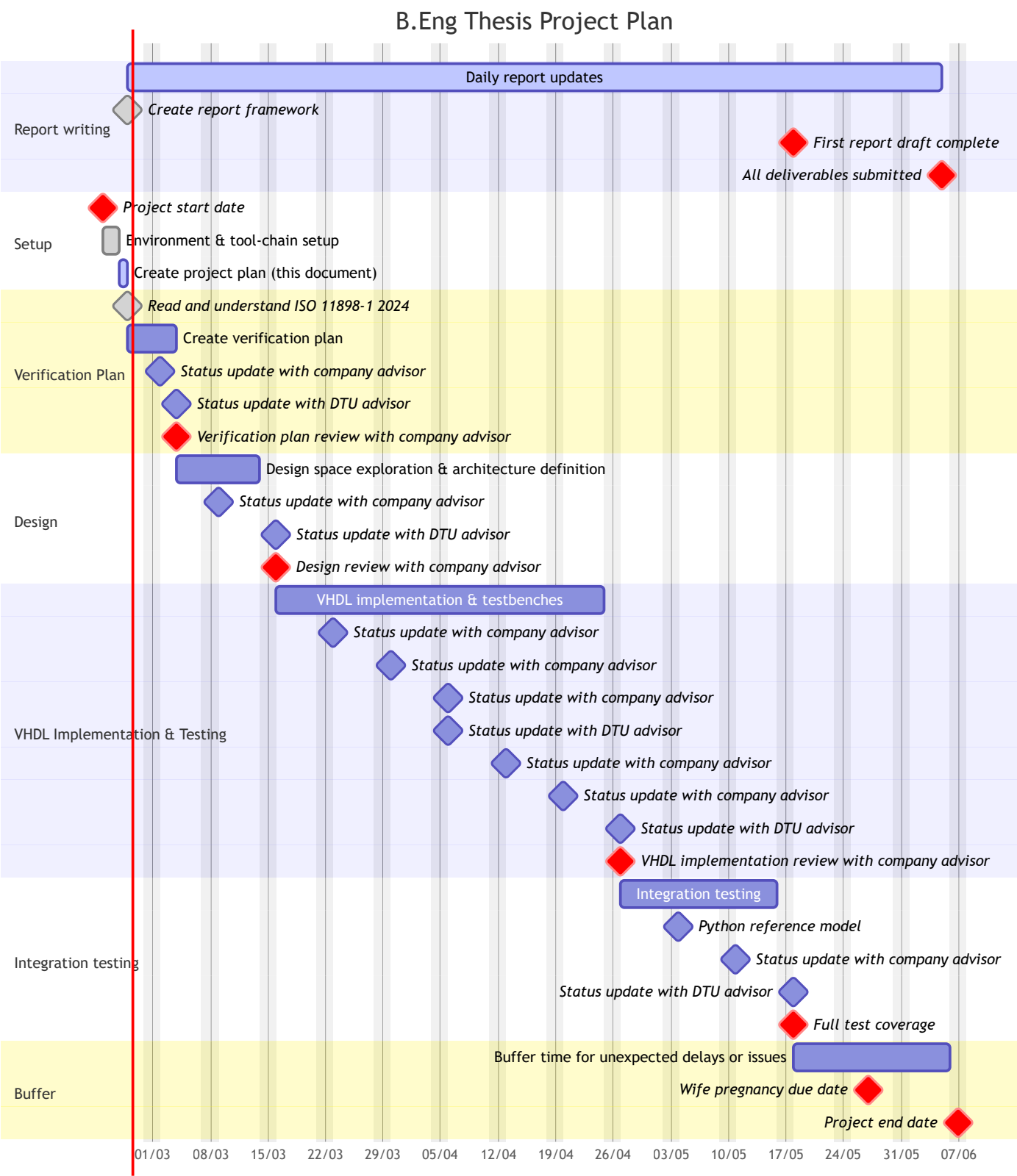
FD CAN bus extension.

The goal of the project is to extend the, by Mads, previously designed CAN bus VHDL module to also include the Flexible Data rate (FD) protocol.

This work will cover:

1. Initial project planning
 2. Read and understand the FD CAN bus standard
 3. Identify updates and how to test these updates
 4. Implementation and testing on a module level
 5. Incorporate the design into the existing engine controller
 6. Write test SW in Python
 7. Document the design
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Gantt Chart



Risks and Mitigation Strategies

Risk	Mitigation
Project schedule slips due to underestimated tasks or unforeseen issues.	Build in buffer time. Review progress weekly. Prioritize critical tasks early.
Report writing takes longer than expected.	Maintain ongoing documentation. Set internal deadlines for drafts. Seek feedback regularly.
Late discovery of design or verification issues.	Develop and follow a detailed verification plan. Run regression tests frequently. Review design with advisors.
Python reference model development takes longer than expected.	Limit reference model to essential features (CRC reference, frame generator, bit stuffing check). Prototype early. Plan for integration testing with hardware and ensure tools are compatible with the target environment.
Hardware integration or testing reveals unexpected issues.	Plan for early integration testing. Coordinate with company advisor for access to hardware and test environments. Document hardware requirements and constraints.
External factors (e.g., family emergencies, advisor availability) impact timeline.	Maintain flexibility. Communicate proactively with stakeholders. Adjust plan as needed.